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vbox_indu@Ubuntu:~$ source bioenv/bin/activate
(bioenv) vbox_indu@Ubuntu:~$ python3
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>> import glob
>>> from Bio import SeqIO
>>> from collections import Counter
>>> import pandas as pd
>>> import os
>>>
>>> file_dir='/home/vbox_indu/Desktop'
```

```
>>> for gfile in gfiles:
...     dir,filename=os.path.split(gfile)
...     filename=filename.strip('.gb')
...     feature_types=read_file(gfile)
...     feature_count=count_features(feature_types)
...     temp_count=[]
...     temp_count.append(filename)
...     for feature in allfeatures:
...         if feature in feature_count.keys():
...             temp_count.append(feature_count[feature])
...         else:
...             temp_count.append(0)
...     allfeature_count.append(temp_count)
...
features have been counted
features have been counted
features have been counted
features have been counted
features have been counted
>>> len(allfeature_count)
5
>>> allfeature_count[0]
['AR465', 59, 2934, 5, 1, 4, 13, 3, 19, 1, 2852]
>>> print(allfeatures)
['tRNA', 'gene', 'misc_feature', 'source', 'misc_binding', 'regulatory', 'ncRNA', 'rRNA', 'tmRNA', 'CDS']
>>> allfeature_count
[['AR465', 59, 2934, 5, 1, 4, 13, 3, 19, 1, 2852], ['M48', 60, 3107, 5, 1, 4, 13, 3, 16, 1, 3027], ['P10', 59, 3250, 0, 1, 0, 10, 3, 8, 1, 3179], ['V521', 60, 3194, 
 1, 3114], ['R50', 59, 3008, 0, 1, 0, 10, 3, 10, 1, 2935]]
>>>
>>> columns=[]
>>> columns.append('File')
>>> columns.extend(allfeatures)
>>> columns
['File', 'tRNA', 'gene', 'misc_feature', 'source', 'misc_binding', 'regulatory', 'ncRNA', 'rRNA', 'tmRNA', 'CDS']
>>>
>>> dataframe=pd.DataFrame(allfeature_count,columns=columns)
>>> print(dataframe)
    File  tRNA  gene  misc_feature  source  misc_binding  regulatory  ncRNA  rRNA  tmRNA   CDS
0  AR465    59  2934             5       1             4          13      3    19      1  2852
1    M48    60  3107             5       1             4          13      3    16      1  3027
2    P10    59  3250             0       1             0          10      3     8      1  3179
3   V521    60  3194             0       1            10           0      3    16      1  3114
4    R50    59  3008             0       1             0          10      3    10      1  2935
>>>
>>> gfiles=glob.glob('%s/*.gb'%file_dir)
>>> gfiles
['/home/vbox_indu/Desktop/AR465.gb', '/home/vbox_indu/Desktop/M48.gb', '/home/vbox_indu/Desktop/P10.gb', '/home/vbox_indu/Desktop/V521.gb', '/home/vbox_indu/Desktop/R50.gb']
>>>
>>> def read_file(gfile):
...     genbank_obj=SeqIO.read(gfile,'gb')
...     features=genbank_obj.features
...     feature_types=[feature.type for feature in features]
...     return feature_types
...
>>> def count_features(feature_types):
...     feature_count=Counter(feature_types)
...     print('features have been counted')
...     return feature_count
...
>>> def scan_all_features(files):
...     allfeatures=[]
...     for gfile in gfiles:
...         feature_types=read_file(gfile)
...         allfeatures.extend(feature_types)
...     allfeatures=set(allfeatures)
...     allfeatures=list(allfeatures)
...     print('all features have been identified')
...     return allfeatures
...
>>> allfeatures=scan_all_features(gfiles)
all features have been identified
>>> print(allfeatures)
['tRNA', 'gene', 'misc_feature', 'source', 'misc_binding', 'regulatory', 'ncRNA', 'rRNA', 'tmRNA', 'CDS']
>>>
>>> allfeature_count=[]
```

```
>>> dataframe=pd.DataFrame(allfeature_count,columns=columns)
>>> print(dataframe)
    File  tRNA  gene  misc_feature  source  misc_binding  regulatory  ncRNA  rRNA  tmRNA   CDS
0  AR465    59  2934             5       1             4          13      3    19      1  2852
1    M48    60  3107             5       1             4          13      3    16      1  3027
2    P10    59  3250             0       1             0          10      3     8      1  3179
3   V521    60  3194             0       1            10           0      3    16      1  3114
4    R50    59  3008             0       1             0          10      3    10      1  2935
>>>
>>> finaldata=dataframe.set_index('File')
>>> finaldata
       tRNA  gene  misc_feature  source  misc_binding  regulatory  ncRNA  rRNA  tmRNA   CDS
File
AR465    59  2934             5       1             4          13      3    19      1  2852
M48      60  3107             5       1             4          13      3    16      1  3027
P10      59  3250             0       1             0          10      3     8      1  3179
V521     60  3194             0       1            10           0      3    16      1  3114
R50      59  3008             0       1             0          10      3    10      1  2935
>>>
>>> columns_to_del=['source']
>>> finaldata.drop(columns=columns_to_del,inplace=True)
>>> finaldata
       tRNA  gene  misc_feature  misc_binding  regulatory  ncRNA  rRNA  tmRNA   CDS
File
AR465    59  2934             5             4          13      3    19      1  2852
M48      60  3107             5             4          13      3    16      1  3027
P10      59  3250             0             0          10      3     8      1  3179
V521     60  3194             0            10           0      3    16      1  3114
R50      59  3008             0             0          10      3    10      1  2935
>>>
>>> finaldata.loc['V521',:]
tRNA              60
gene            3194
misc_feature       0
misc_binding      10
regulatory         0
ncRNA              3
rRNA              16
tmRNA              1
CDS             3114
Name: V521, dtype: int64
>>> #these process is to query the dataframe
>>>
>>> finaldata.loc['V521','gene']
np.int64(3194)
>>>
>>> outputfile='/home/vbox_indu/Desktop/featurecount.csv'
>>> finaldata.to_csv(outputfile,index=True)
>>>
>>>
```